🛒 Zepto E-commerce SQL Data Analyst Portfolio Project

This is a complete, real-world data analyst portfolio project based on an e-commerce inventory dataset scraped from Zepto — one of India's fastest-growing quick-commerce startups. This project simulates real analyst workflows, from raw data exploration to business-focused data analysis.

This project is perfect for:

- 📊 Data Analyst aspirants who want to build a strong **Portfolio Project** for interviews and LinkedIn
- 📚 Anyone learning SQL hands-on
- 💼 Preparing for interviews in retail, e-commerce, or product analytics

📌 Project Overview

The goal is to simulate how actual data analysts in the e-commerce or retail industries work behind the scenes to use SQL to:

✅ Set up a messy, real-world e-commerce inventory **database**

✅ Perform **Exploratory Data Analysis (EDA)** to explore product categories, availability, and pricing inconsistencies

✅ Implement **Data Cleaning** to handle null values, remove invalid entries, and convert pricing from paise to rupees

✅ Write **business-driven SQL queries** to derive insights around **pricing, inventory, stock availability, revenue** and more

📁 Dataset Overview

The dataset was sourced from Kaggle and was originally scraped from Zepto's official product listings. It mimics what you'd typically encounter in a real-world e-commerce inventory system.

Each row represents a unique SKU (Stock Keeping Unit) for a product. Duplicate product names exist because the same product may appear multiple times in different package sizes, weights, discounts, or categories to improve visibility – exactly how real catalog data looks.

📄 Columns:

- **sku_id:** Unique identifier for each product entry (Synthetic Primary Key)
- **name:** Product name as it appears on the app
- **category:** Product category like Fruits, Snacks, Beverages, etc.
- **mrp:** Maximum Retail Price (originally in paise, converted to ₹)
- **discountPercent:** Discount applied on MRP
- **discountedSellingPrice:** Final price after discount (also converted to ₹)
- **availableQuantity:** Units available in inventory
- **weightInGms:** Product weight in grams
- **outOfStock:** Boolean flag indicating stock availability

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- **outOfStock:** Boolean flag indicating stock availability
- **quantity:** Number of units per package (mixed with grams for loose produce)

🔧 Project Workflow

Here's a step-by-step breakdown of what we do in this project:

1. Database & Table Creation

We start by creating a SQL table with appropriate data types:

```
CREATE TABLE zepto (
  sku_id SERIAL PRIMARY KEY,
  category VARCHAR(120),
  name VARCHAR(150) NOT NULL,
  mrp NUMERIC(8,2),
  discountPercent NUMERIC(5,2),
  availableQuantity INTEGER,
  discountedSellingPrice NUMERIC(8,2),
  weightInGms INTEGER,
  outOfStock BOOLEAN,
  quantity INTEGER
);
```

2. Data Import

- Loaded CSV using pgAdmin's import feature.
- If you're not able to use the import feature, write this code instead:

```
  \copy zepto(category,name,mrp,discountPercent,availableQuantity,
          discountedSellingPrice,weightInGms,outOfStock,quantity)
FROM 'data/zepto_v2.csv' WITH (FORMAT csv, HEADER true, DELIMITER ',', QUOTE '"', ENCODING 'UTF8');
```

- Faced encoding issues (UTF-8 error), which were fixed by saving the CSV file using CSV UTF-8 format.

3. 🔍 Data Exploration

- Counted the total number of records in the dataset
- Viewed a sample of the dataset to understand structure and content
- Checked for null values across all columns
- Identified distinct product categories available in the dataset
- Compared in-stock vs out-of-stock product counts
- Detected products present multiple times, representing different SKUs

4. 🧹 Data Cleaning

- Identified and removed rows where MRP or discounted selling price was zero
- Converted mrp and discountedSellingPrice from paise to rupees for consistency and readability

5. 📊 Business Insights

- Found top 10 best-value products based on discount percentage
- Identified high-MRP products that are currently out of stock
- Estimated potential revenue for each product category
- Filtered expensive products (MRP > ₹500) with minimal discount
- Ranked top 5 categories offering highest average discounts
- Calculated price per gram to identify value-for-money products
- Grouped products based on weight into Low, Medium, and Bulk categories
- Measured total inventory weight per product category

[10]:
```python
import pandas as pd

df = pd.read_csv("zepto_v2.csv", encoding="latin1")
```

[3]:
```python
df.head(5)
```

[3]:

	Category	name	mrp	discountPercent	availableQuantity	discountedSellingPrice	weightInGms	outOfStock	quantity
0	Fruits & Vegetables	Onion	2500	16	3	2100	1000	False	1
1	Fruits & Vegetables	Tomato Hybrid	4200	16	3	3500	1000	False	1
2	Fruits & Vegetables	Tender Coconut	5100	15	3	4300	58	False	1
3	Fruits & Vegetables	Coriander Leaves	2000	15	3	1700	100	False	100
4	Fruits & Vegetables	Ladies Finger	1400	14	3	1200	250	False	250

[4]:
```python
!pip install psycopg2-binary sqlalchemy
```

```
Requirement already satisfied: psycopg2-binary in c:\python313\lib\site-packages (2.9.11)
Requirement already satisfied: sqlalchemy in c:\python313\lib\site-packages (2.0.48)
Requirement already satisfied: greenlet>=1 in c:\python313\lib\site-packages (from sqlalchemy) (3.3.2)
Requirement already satisfied: typing-extensions>=4.6.0 in c:\python313\lib\site-packages (from sqlalchemy) (4.15.0)

[notice] A new release of pip is available: 25.1.1 -> 26.0.1
[notice] To update, run: python.exe -m pip install --upgrade pip
```


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```


[6]:
```python
from sqlalchemy import create_engine
```

[7]:
```python
from sqlalchemy import create_engine

# MySQL connection
engine_mysql = create_engine(
    "mysql+pymysql://root:Naruto%408652@localhost:3306/zepto"
)

try:
    engine_mysql.connect()
    print("MySQL Connected Successfully")
except Exception as e:
    print("Connection failed:", e)
```

```
MySQL Connected Successfully
```

[8]:
```python
import pandas as pd
```

[11]:
```python
df.to_sql(name='zepto_v2', con=engine_mysql, if_exists='append', index=False)
```

[11]:
```
3732
```

zepto_SQL_project/postgres@PostgreSQL 17*

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No limit

Query Query History

```sql
create table zepto (
sku_id SERIAL PRIMARY KEY,
category VARCHAR(120),
name VARCHAR(150) NOT NULL,
mrp NUMERIC(8,2),
discountPercent NUMERIC(5,2),
availableQuantity INTEGER,
discountedSellingPrice NUMERIC(8,2),
weightInGms INTEGER,
outOfStock BOOLEAN,
quantity INTEGER
);
```

zepto_SQL_project/postgres@PostgreSQL 17

No limit

Query Query History

```sql
quantity INTEGER
);

--data exploration

--count of rows
select count(*) from zepto;

--sample data
SELECT * FROM zepto
LIMIT 10;

--null values
SELECT * FROM zepto
```

Data Output Messages Notifications

	sku_id [PK] integer	category character varying (120)	name character varying (150)	mrp numeric (8,2)	discountpercent numeric (5,2)	availablequantity integer	discountedsellingprice numeric (8,2)	weightingms integer	outofst boolear
1	1	Fruits & Vegetables	Onion	25.00	16.00	3	21.00	1000	false
2	2	Fruits & Vegetables	Tomato Hybrid	42.00	16.00	3	35.00	1000	false
3	768	Munchies	Everest Sabji Masala	30.00	3.00	6	29.00	50	false
4	842	Munchies	Popular Essentials Ajwain Wh...	50.00	12.00	2	44.00	100	false
5	1272	Beverages	iD Curd Cup	60.00	0.00	6	60.00	400	false
6	1543	Packaged Food	Kellogg's Corn Flakes	425.00	9.00	1	383.00	1000	false

Total rows: 10 of 10 Query complete 00:00:00.215 Ln 23, Col 10

No limit

Query | Query History

```sql
--different product categories
SELECT DISTINCT category
FROM zepto
ORDER BY category;

--products in stock vs out of stock
SELECT outOfStock, COUNT(sku_id)
FROM zepto
GROUP BY outOfStock;

--product names present multiple times
SELECT name, COUNT(sku_id) AS "Number of SKUs"
FROM zepto
```

Data Output | Messages | Notifications

	category character varying (120)
1	Beverages
2	Biscuits
3	Chocolates & Candies
4	Cooking Essentials
5	Dairy, Bread & Batter

zepto_SQL_project/postgres@PostgreSQL 17

No limit

Query | Query History

```sql

-- Q1. Find the top 10 best-value products based on the discount percentage.
SELECT DISTINCT name, mrp, discountPercent
FROM zepto
ORDER BY discountPercent DESC
LIMIT 10;

--Q2.What are the Products with High MRP but Out of Stock

SELECT DISTINCT name,mrp
FROM zepto
WHERE outOfStock = TRUE and mrp > 300
ORDER BY mrp DESC;
```

Data Output | Messages | Notifications

	name character varying (150)	mrp numeric (8,2)	discountpercent numeric (5,2)
1	Dukes Waffy Orange Wafers	45.00	51.00
2	Dukes Waffy Chocolate Wafers	45.00	51.00
3	Dukes Waffy Strawberry Wafers	45.00	51.00
4	RRO Mozzarella Block Cheese	295.00	50.00
5	Ceres Foods Laal Maas Instant Liquid Masala	220.00	50.00
6	Chef's Basket Durum Wheat Elbow Pasta	160.00	50.00

Total rows: 10 of 10 Query complete 00:00:00.106 Ln 84, Col 10

```sql
--Q2.What are the Products with High MRP but Out of Stock

SELECT DISTINCT name,mrp
FROM zepto
WHERE outOfStock = TRUE and mrp > 300
ORDER BY mrp DESC;

--Q3.Calculate Estimated Revenue for each category
SELECT category,
SUM(discountedSellingPrice * availableQuantity) AS total_revenue
FROM zepto
GROUP BY category
```

	category character varying (120)	total_revenue numeric
1	Fruits & Vegetables	10846.00
2	Meats, Fish & Eggs	20693.00
3	Biscuits	25007.60
4	Beverages	55051.00
5	Dairy, Bread & Batter	55051.00
6	Health & Hygiene	64180.00

Total rows: 14 of 14 Query complete 00:00:00.207 Ln 95, Col 14

```sql
-- Q6. Find the price per gram for products above 100g and sort by best value.
SELECT DISTINCT name, weightInGms, discountedSellingPrice,
ROUND(discountedSellingPrice/weightInGms,2) AS price_per_gram
FROM zepto
WHERE weightInGms >= 100
ORDER BY price_per_gram;

--Q7.Group the products into categories like Low, Medium, Bulk.
SELECT DISTINCT name, weightInGms,
CASE WHEN weightInGms < 1000 THEN 'Low'
    WHEN weightInGms < 5000 THEN 'Medium'
    ELSE 'Bulk'
```

	name character varying (150)	weightingms integer	discountedsellingprice numeric (8,2)	price_per_gram numeric
1	Tata Salt	1000	24.00	0.02
2	Onion	1000	21.00	0.02
3	Onion	3000	57.00	0.02
4	Shubh kart - Nirmal sugandhi mogra wet dhoop zipper 20 sticks	1160	28.00	0.02
5	Vicks Cough Drops Menthol	1160	20.00	0.02
6	Aashirvaad Iodised Salt	1000	19.00	0.02

Total rows: 1000 of 1395 Query complete 00:00:00.219 Ln 116, Col 62